



Multi-frame feature-fusion-based model for violence detection

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Published online: 24 June 2020
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Abstract

Human behavior detection is essential for public safety and monitoring. However, in human-based surveillance systems, it requires continuous human attention and observation, which is a difficult task. Detection of violent human behavior using autonomous surveillance systems is of critical importance for uninterrupted video surveillance. In this paper, we propose a novel method to detect fights or violent actions based on learning both the spatial and temporal features from equally spaced sequential frames of a video. Multi-level features for two sequential frames, extracted from the convolutional neural network's top and bottom layers, are combined using the proposed feature fusion method to take into account the motion information. We also proposed Wide-Dense Residual Block to learn these combined spatial features from the two input frames. These learned features are then concatenated and fed to long short-term memory units for capturing temporal dependencies. The feature fusion method and use of additional wide-dense residual blocks enable the network to learn combined features from the input frames effectively and yields better accuracy results. Experimental results evaluated on four publicly available datasets: HockeyFight, Movies, ViolentFlow and BEHAVE show the superior performance of the proposed model in comparison with the state-of-the-art methods.

Keywords Violence detection · Autonomous Video Surveillance · CNN-LSTM · Feature fusion · Spatio-temporal features

1 Introduction

The applications of surveillance cameras are rapidly becoming essential for monitoring important places and to observe offensive activities like robbery, theft, public violence, fights, and many other criminal acts. The monitoring system, therefore, has to be made effective to observe and control these events on a regular basis. The continuity in surveillance is a significant factor; however, it is difficult and not humanly feasible to monitor these activities and view surveillance footages continuously and without interruption due to less human resources. As abnormal activities occur infrequently and the probability of occurrence of violent behavior is minimal. Therefore, an efficient autonomous monitoring system can do the needful task which possesses the ability to detect

the abnormal and violent activities independent of human supervision. There are difficulties in identifying offensive actions, for example, an action of a person may look aggressive, actually being normal and vice versa. The subjective nature of the problem may cause the model to misclassify the results. Also, the surveillance videos acquired from CCTV cameras, are mostly of low resolution and sometimes lack minute helpful details required to identify violent actions. Human violence detection is based on the related computer vision tasks such as object detection, tracking, action recognition, and video classification. For object detection and classification from static images, only the spatial information is required to identify or classify objects in images. As videos consist of sequence of static images or frames, where every action is represented by both spatial and motion information throughout the duration of a specific action. Therefore, it is essential to incorporate both spatial and temporal features to classify a particular action in a video correctly. To utilize temporal information, many methods have been proposed [1–6] for recognition of a specific action. The recent development in deep learning-based architectures has helped in achieving state-of-the-art results in many computer vision-related areas. However, the progress for the detection of violent

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human behavior is slow. A deep-learning-based model can effectively extract features from the data without the need for complex data pre-processing. In contrast to handcrafted-based models, the deep architectures can learn image features from raw pixel inputs. Therefore, similar architectures can be used for several related tasks by slightly modifying the architecture.

In this paper, we present a novel deep-learning-based binary classification architecture for human violence detection. The proposed model can effectively learn appearance and motion features from fused feature maps using convolutional and recurrent neural networks. The main contributions of our proposed method are as follow:

- We proposed a novel multi-level feature fusion method to incorporate local motion patterns using features from sequential pair of input frames.
- Wide-Dense Residual Block (WDRB) is proposed which is used to learn multi-level combined fused feature maps, that result in increased overall frame-level detection accuracy.
- The proposed method results in the prediction for each input frame instead of video level prediction. The experimental results on various standard benchmark datasets show the effectiveness of the proposed method compared to the existing methods.

The rest of the paper is organized as follows: Section 2 discusses the related work in the field. Section 3 explains the details of proposed architecture. Section 4 provides the implementation details of the proposed methods. Section 5 discuss the overview of datasets, experimental results and a comparison with the existing state-of-the-art methods. At the end, the Sect. 6 concludes this paper.

2 Related work

Various methods have been proposed for human violence detection in videos. Some earlier techniques use only the audio features [7] to detect violent actions. Other methods such as [8–10] are based on both audio and visual features which gained significant importance in the detection of different human actions. In [11] the authors used the presence of fire or blood in the video, along with sound characteristics to categorize a particular violent action. The use of both audio and visual features may often be more favorable; however, in most of the surveillance footage the audio information is not available. Therefore, we address the problem of violence detection, which focuses only on visual features. Moreover, the identification of violence is of broad importance, involving not just person to person violence, but also sports, crowd and physical violent actions. The existing

methods for violence detection can be categorized into two classes. First, which use rapid inter-frame motion variation patterns between the videos frame to detect fighting action [12]. Second, in which the change of motion is analyzed or tracked in the video [1,2,12–16]. Many earlier methods are based on handcrafted features where these features are manually extracted and a learning model is used to learn the local features and the deviations from the learned model are considered as outliers or abnormal events. Local features such as Histogram of Oriented Flows (HOF) [17] Space Time Interest Points (STIP) [18,19], Histogram of Oriented Gradients [20], Motion Scale Invariant Feature Transform (MoSIFT) [21] are first introduced for action recognition. These methods use limbs position and motion tracking techniques to find the interest points related to spatiotemporal features [14,15]. Zhang et al. in [13] proposed improved MoIWL feature descriptor by using appearance and motion information in Weber local descriptor. In [2], the authors proposed a new descriptor named violent flow (ViF) for detection of crowd violence in videos which is based on variations of optical flow magnitudes between subsequent video frames. In [22] Gao et al. improved the ViF by proposing Oriented Violent Flow (OVIF) which adds orientation information in flow magnitude feature vectors. In recent times, with the advancement of deep learning based architectures, many methods have been proposed for computer vision-related tasks such as image classification [23,24], action recognition [3,4,25,26], human emotion recognition [8,27], anomaly detection [28–30], and medical image analysis [31,32] etc. Simonyan and Zisserman [4] proposed a two stream architecture using two separate convolutional neural networks (CNN) for spatial features and optical flow respectively to learn both spatiotemporal features effectively by fusing information from both CNNs over multiple frames. Tran et al. [5] introduced 3D CNN for action recognition which can learn both appearance and motion features from a short video clip using a single end-to-end architecture. In [6] Donahue et al. proposed long term recurrent convolutional network (LRCN) for image captioning, activity recognition and video description. This LRCN architecture is based on combining long short-term memory (LSTM) units and CNNs to effectively learn long term dependencies in input video sequences. In [33] ConvLSTM model is proposed for precipitation nowcasting, where the authors replaced the CNNs fully connected (FC) layers with LSTMs and FC (FC-LSTMs) layers. In [34], the authors used a similar model based on CNN and LSTMs for the purpose of violent action detection in videos. Recently, Hanson et al. [35] proposed a violence detection model by using bi-directional LSTM instead of simple LSTMs along with CNN for the purpose of violence detection.

This paper presents a novel method for violence detection in videos based on CNN and LSTM with the addition of proposed Wide-Dense Residual blocks (WDRB). The

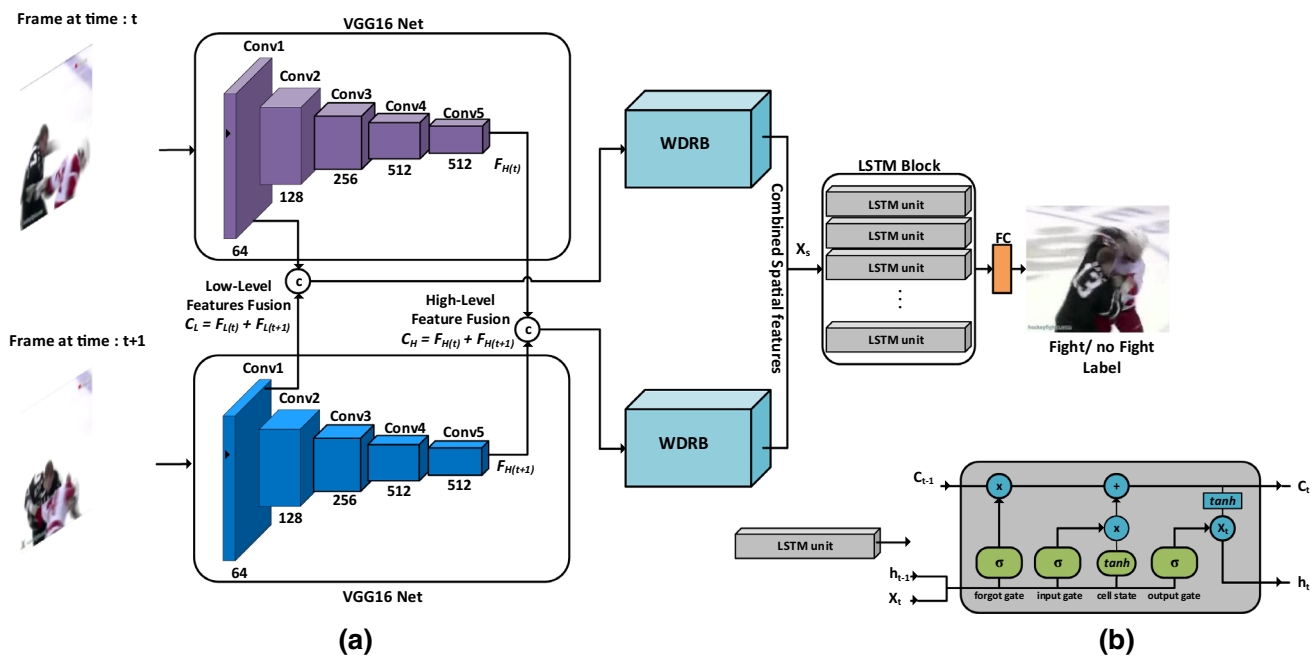


Fig. 1 **a** The proposed violence detection model: The model takes inputs from two video frames at time t and $t + 1$ and the pre-trained CNNs are used to extract both high and low-level features. The additional Wide Dense Residual blocks (WDRB) are used to learn these

combined feature maps of the two frames, and then the LSTM network learns the temporal patterns between the frame features. **b** Structure of LSTM cell unit

proposed method can effectively learn the multi-level inter-frame dependency features using feature fusion method. The proposed method has the ability to detect violent actions from variable length videos while predicting the results at frame-level. Fig. 1 shows the main architecture of the proposed model.

2.1 Motivation

The increase in criminal happenings, their suddenness, and the resultant possibility of inflicted harm spurred law enforcement and government agencies to use comprehensive security surveillance systems to identify violent interactions. However, autonomous detection of violence still remains an unsolved issue due to specific features and complex nature of the problem. The difficulties in detecting violent actions in surveillance videos are subjected to many factors such as, the quality of video footage, lightning conditions, and no contextual information to ease the detection of violent or non-violent behavior. One of the difficulty in human violence detection is to recognize a violent action that is happening instantaneously for a short duration of time. Most of the existing violence detection models result in the prediction on clip-level instead of frame-level prediction where some frames may not include violent activity. Following the intuition of recent works proposed in [34–36] that CNN architecture followed by LSTM can well learn both spa-

tial and temporal features from the videos. Motivated from this, we aim to improve the CNN-LSTM model for violence detection by incorporating multi-level feature fusion along with addition of proposed wide dense residual layer blocks to predict the violent activity for each input frame that provides better performance results for videos which consist of violent actions for a very short duration.

3 Proposed methodology

This paper proposes a novel supervised binary classification method to differentiate between violent and non-violent activities in videos using an end-to-end trainable model which combines CNN and LSTM networks to learn both spatial and temporal features from a sequence of input frames to classify the videos into both categories. To learn local temporal features, it is essential to incorporate motion information. Most of the existing techniques [2,17,22,37] rely on optical flow method for temporal information. Optical flow is usually calculated through the difference of consecutive frames in the regions where motion or movement of the object is detected. Some previous methods [17,20] used handcrafted features for computing optical flows, whereas some recent methods [38] used deep CNNs for learning optical flows. In this paper, instead of using optical flow as motion features, two adjacent video frames containing different motion pat-

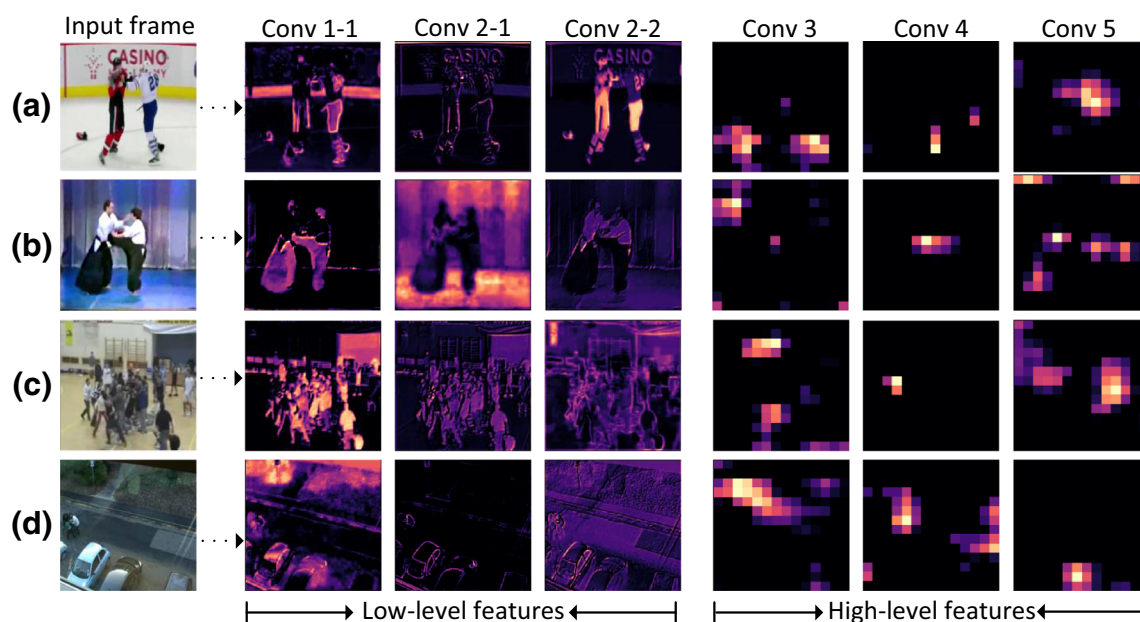


Fig. 2 Visualization of low-level and high-level feature map from different layers of CNN model using samples from four datasets. **a** Hockey-Fight dataset sample feature maps **b** Movies **c** Crowd Violence and **d** BEHAVE dataset sample feature maps from fighting scenes

terns corresponding to a specific action (taken at time-stamp t and $t + 1$ respectively) are used as input to the network. In order to learn spatial features from the input image frames, the network takes advantages of transfer learning and uses two pre-trained VGG-16 [39] networks on the ImageNet [40] dataset for extracting both low-level and high-level spatial features from two consecutive input frames. Then these low-level and high-level features are combined using the feature fusion method and then combined feature maps are learned using proposed wide-dense residual layer block that leads to improved overall network performance accuracy. The following Sects. 3.1 and 3.2 describes the feature learning procedure in detail.

3.1 Spatial feature learning using feature fusion method

Deep CNNs have been widely used for learning image structural and semantic contextual information as they can effectively learn the high-level features. Different layers of convolutional neural network incorporate multiple image features. Most of the existing algorithms [4,6,23] use CNN's top layer or high-level features for the spatial information as these layers are based on semantic information. However, the low-level image features can represent object localization and contain more detailed information more specifically as they are closer to the inputs. Fig. 2 shows visualization example of both low and high-level feature maps from different layers of CNN network for samples of four datasets used in experimental evaluation. Many recent methods, such as

[41,42] used multiple layer features by combining both low-level and high-level features that can significantly improve the performance accuracy of the network. In [42], the U-net architecture combines the low-level features with high-level features and obtained a good accuracy on biomedical image segmentation. Our technique for fusing multi-level spatial feature is different from [3,41,42]. In [3] the early fusion method combines information in a specified time window by extending convolution filter in temporal dimension by T which results in a filter size of $(11 \times 11 \times 3 \times T)$, where T is the number of frames in temporal dimension. In late fusion, two separate networks of single frame baseline models share the network parameter for a temporal window of 15 frames and then concatenates the features of two networks in first fully-connected layer in order to learn global motion features.

However, the proposed method learns the spatial features from two sequential input frames at time t and $t + 1$ and for each of these two input frames, both high-level and low-level features are extracted using a VGG-16 network which is pre-trained on ImageNet [40] dataset. After feature extraction, the bottom layer low-level feature maps of the frame at time t are combined with low-level features of the frame at time $t + 1$ to learn local motion patterns. Similarly, the high-level features at the top layer for the frame at time t are fused with high-level features of frame $t + 1$ to learn global motion characteristics where t is current time-stamp of input frame. Let $F_{L(t)}, F_{L(t+1)} \in \mathbb{R}^{h_L \times w_L \times d_L}$ denote the low-level feature maps of frames at time t and $t + 1$ and $F_{H(t)}, F_{H(t+1)} \in \mathbb{R}^{h_H \times w_H \times d_H}$ represents high-level feature maps of frames at

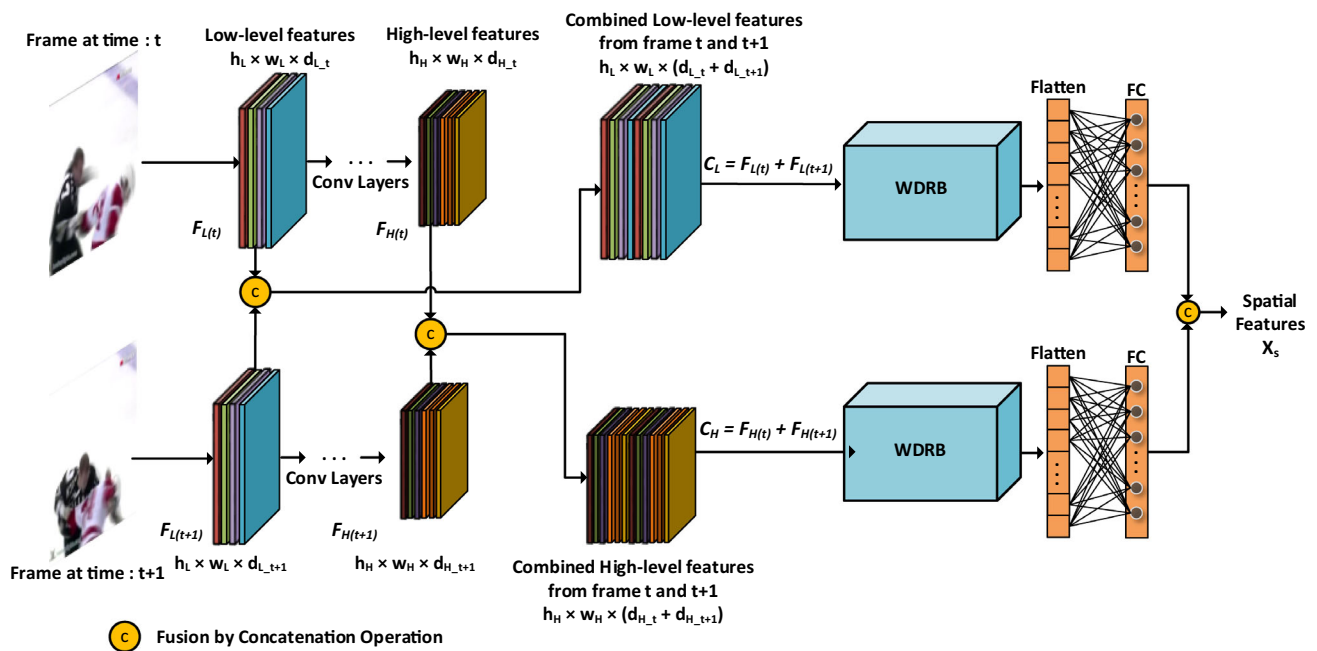


Fig. 3 Spatial feature learning through feature fusion of two sequential input frames (at time t and $t + 1$). The network output a combined spatial feature vector X_s by fusing both low-level and high-level features from input frame pair

time t and $t + 1$ respectively. Here h and w represent the height and width, and d is the channel depth, where the subscripts L and H differentiate between low and high-level features. The low-level feature fusion is described by the Eq. 1 and the high-level features are fused by using Eq. 2

$$C_L = F_{L(t)} + F_{L(t+1)} \quad (1)$$

$$C_H = F_{H(t)} + F_{H(t+1)} \quad (2)$$

where the '+' operator represents the fusion of feature maps by concatenation. Figure 3 shows the spatial feature learning model based on fused feature maps.

As the features are extracted using a pre-trained VGG-16 network where the input to the network is a fixed size image of size 224×224 . The VGG-16 has a total of 5 stacks of convolutional layers with different depths where the top convolutional layer, from which the low-level features F_L are extracted, has the size of $224 \times 224 \times 64$ and the bottom layer where the high-level feature maps F_H has the size of $14 \times 14 \times 512$. Using Eqs. 1 and 2 the size of combined low-level feature maps C_L for two input frames is $224 \times 224 \times 128$, whereas the size of C_H , the combined high-level features is $14 \times 14 \times 1024$. The pooling and fully connected layers of the VGG-16 network is detached, and an additional CNN block is added to the network which is based on proposed wide-dense residual convolutional layers to add more depth to the network without degrading the input size. Adding more depth using the residual layer module can increase the accuracy of the network [43] and as each module is added as identity

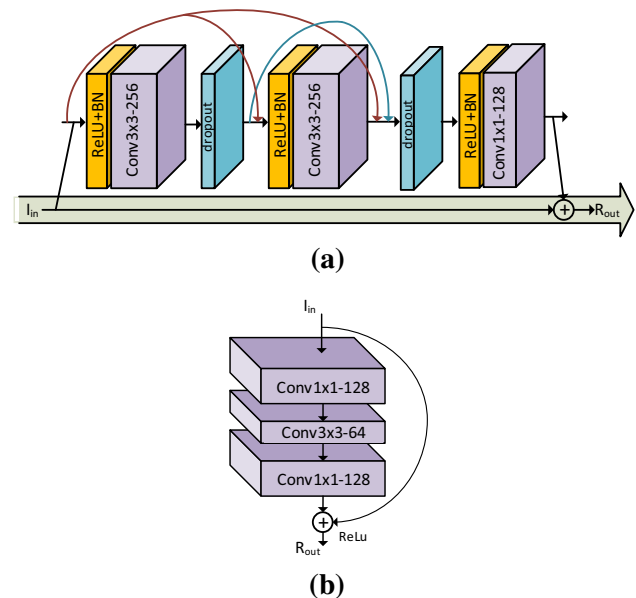


Fig. 4 **a** Architecture of Proposed Wide-Dense Residual Block (WDRB) and **b** Simple Residual Block [43]

mapping function, so that it does not increase the training error.

These additional layers learn the combined motion patterns from concatenated feature maps of two frames. The WDRB shown in Fig. 4a is based on the stack of three convolutional layers, where the first two layers has total of 512 filters of size 3×3 , whereas the last layer in the block has a

filter size of 1×1 with a total 256 filters to downsample the feature maps. Compared to the original residual block [43] shown in Fig. 4b the proposed block consist of widened convolutional layers which results in computationally efficient performance [44]. Apart from widening the layers, we also densely connected the first two layers in Wide Dense Residual Block (WDRB) to exploit the hierarchical features from both convolutional layers because the densely connected layers can better utilize all the hierarchical feature between the layers [45]. Moreover for each layer in the WDRB, rectified linear unit(ReLU) and batch normalization operations are applied before each convolution operation and the dropout with a probability of 0.3 is added after each layer in the proposed residual block.

The use of dropout is first introduced by [46] where it is used to prevent overfitting and co-adaption due to large number of parameters in top layers. The authors in [46] reported a gain in performance of up to 2.55% to 3.02% by using dropout in convolntional layers. The dropout is generally substituted with batch normalization [47] operation which acts as a regularizer to minimize the internal covariate-shift by normalizing the network activation. In our proposed WDRB, we used both dropout and batch-normalization. As in our case, widened convolutional layers with dense connections result in an increase number of parameters. Previously, He et al [43] used dropout in identity mappings for residual block. However, we experimented dropout after the convolution layers of our WDRB which yields better results compared to simple residual block.

The residual block is defined in Eq. 3.

$$R = \mathcal{F}(I, W_i) + I \quad (3)$$

Here R is the output vector from the residual block, I is the input and $\mathcal{F}(I, W_i)$ is the residual identity mapping function that is learned by additional residual block. Here, $I = C_L$ when the input to residual block is combined low-level features and $I = C_H$ for combined high-level feature maps. The output from each residual layer block can be represented by Eqs. 4 and 5.

$$R_1 = \mathcal{F}(C_L, \{W_{C_L}\}) + C_L \quad (4)$$

$$R_2 = \mathcal{F}(C_H, \{W_{C_H}\}) + C_H \quad (5)$$

The output of residual blocks is then passed to a fully connected layer, to flatten the output of convolutional layer into one dimensional vector, that has a total of 1024 neurons. The output of fully connected layers for both frames are then concatenated that forms the combined spatial features X_s of both input frames.

3.2 Temporal feature learning through LSTM

As videos are composed of multiple frames and each action in a video is described by the motion information in multiple frame sequence. For learning temporal dependencies, simple CNNs cannot provide the best result as they do on the image classification task. LSTMs combined with CNNs can efficiently learn both temporal dependencies [36] and spatial features in the sequence of multiple input frames and make accurate predictions for the tasks of activity recognition, image captioning and video description. In order to learn temporal features as well as global motion patterns between the two frames, the combined fixed length spatial feature vector X_s is fed to the LSTM. The combined feature X_s is based on combined high and low-level image feature maps of two frames C_L and C_H . The ability of the LSTM to learn long temporal dependencies from the CNN feature vectors [6] allows the model to learn the internal representation of motion information over time from sequential input data. The general LSTM cell has parameters W , b , and it maps the input X_s and previous hidden state h_{t-1} to the output c_t and also update the current state h_t , as shown in Fig. 1b.

The LSTM layer make a final prediction by averaging label probabilities $y_t \in \theta$, for each pair of input frame sequence using a softmax classifier, where θ is a finite set of outcomes. The prediction distribution $P(\mathbf{Y}_t)$ is defined in Eq. 6

$$P(\mathbf{Y}_t = \theta) = \text{softmax}(\hat{y}_t) = \frac{e^{\hat{y}_t(\theta)}}{\sum_{\theta'=0}^k e^{\hat{y}_t(\theta')}} \quad (6)$$

where $\theta' \in \theta$, $\hat{y}_t = W_t c_t + b_t$ is the liner prediction layer, W and b are the trained LSTM parameters, c_t is the output which is passed through prediction layer $\hat{y}_t$ and t is current time stamp.

3.3 Model optimization

In case of detecting violent actions from surveillance videos, the positive detection case ($t = 1$) corresponds to fighting action and the negative case ($t = 0$) is a non-fighting or normal action. Considering the reliability and simplicity of the logistic regression function for binary classification applications, in this paper we use a sigmoid-based cross-entropy loss function for classification of violent actions. First, the hidden states from LSTMs are flattened to a 1D vector which is then used to predict the violent behavior. The following Eq. 7 shows the cross-entropy loss function.

$$\text{loss} = \frac{1}{N} \sum_{i=1}^{M=2} \sum_{j=1}^N [-t_{ij} \log(p_{ij}) + (1 - t_{ij}) \log(1 - p_{ij})] \quad (7)$$

where $p_j = \text{sigmoid}(W.h_j + b_j)$ is output prediction probability for j^{th} sample, N is the total number of training samples, M is prediction classes (i.e. fight or no-fight), t represents the ground truth values (i.e. $t = 0$ if the class is non-fight and $t = 1$ for fighting category)

4 Implementation details

4.1 Pre-processing

The dataset is first divided into two categories of violent and non-violent videos and equally spaced frames from each video are extracted and resized to a size of 224×224 pixels. Each input frame values are normalized in the range of 0 to 1. For each training video, maximum of f_N frames are selected and if a video has more than f_N frames, than frames are selected at regular intervals by skipping intermediate frame to avoid redundant computations. The datasets used in experimental results, all have video lengths in the range of 1 to 5 s and with a frame rate of 25fps. The value $f_N = 30$ is selected for all these dataset and frames for training will be picked by skipping each intermediate frame. For example, for a video sample with total of 60 frames, only 30 frames are picked for training sample by omitting each intermediate frame. The value of f_N is selected on the basis of video length in datasets and If the video length is larger, then the value of f_n will be selected accordingly. Apart from converting the video to frames and resizing each frame, no further frame pre-processing is needed. Moreover, because of limited amount of training samples, data augmentation technique is used for generating extra training samples. Different augmentation types including random frame cropping of size 224×224 , horizontal flipping and translation transforms are used for HockeyFight, Movies and CrowdViolence datasets. However, for the BEHAVE dataset only horizontal flipping is used.

4.2 Experimental setup

For spatial feature extraction pre-trained weights are used for VGG-16 and only the additional WDRB and the LSTM units are trained on input data whereas the pre-trained model is not trained during the training phase. The fully connected layers of pre-trained CNNs are replaced with an additional residual layer blocks. Each residual block used in the proposed work contains a stack of three convolutional layers. The first layer of proposed residual block have filter size 3×3 with total 256 filters. The second layer also contains 256 filters of size 3×3 and the third layer in residual block have 128 filter of size 1×1 . As recurrent neural networks take more time to converge because of the sequential inputs and accumulation of gradients which causes the loss to propagate between the

intermediate states. The gradient starts to fluctuate and result in an exploding gradient problem, Therefore, gradient clipping method [49] is used to solve this problem. The dropout [50] with a probability of 0.2 is also used between the LSTM cells. The LSTM layer has total of 512 units The batch for the model has shape (b, f_N, w, h, d) where b is batch size, f_N is number of frames in a video and w, h and d are the width, height, and depth of input respectively. The batch size $b = 5$ videos is selected where each video has a maximum of $f_N = 30$ frames. The initial learning rate of 10^{-3} is used, and the network is trained for 100 epochs for each training set. The ADAM optimization function is used for backpropagation to minimize the cross-entropy loss shown in Eq. 7. The momentum decay rate values of 0.9 and 0.999 are used. The network is trained using Nvidia GTX Titan-X GPUs and the model is implemented in Python¹ using tensorflow framework².

4.3 Video-level accuracy calculation

As the network is trained using the sequential pair of input frames of a video, and for each input the model predicts the violence category. However, the existing methods evaluate the model accuracy at video-level. In order to compare the accuracy evaluation of proposed method with existing approaches, the continuous frame-level predictions for a specific class are accumulated and if the accumulated prediction is larger than a certain threshold value then the clip-level prediction is generated for that particular class.

The threshold value is obtained by continuously accumulating frame-level predictions up to the length of the training video i.e. $f_N = 30$ (for our experiments) and looking for a threshold value that gives maximum accuracy value. The Fig. 5 shows the results of accuracy performance for different values of threshold where the small threshold value ranging from 5 to 10 yields maximum performance. Therefore, for our network we chose the threshold value of 5 frames.

5 Experimental details

In this section, we discuss the dataset overview and the detailed analysis of experimental results. In order to show the effectiveness of proposed method, the comparative analysis with several state-of-the-art techniques is also presented which includes both deep learning-based and handcrafted feature-based methods.

¹ <https://www.python.org>.

² <https://www.tensorflow.org>.

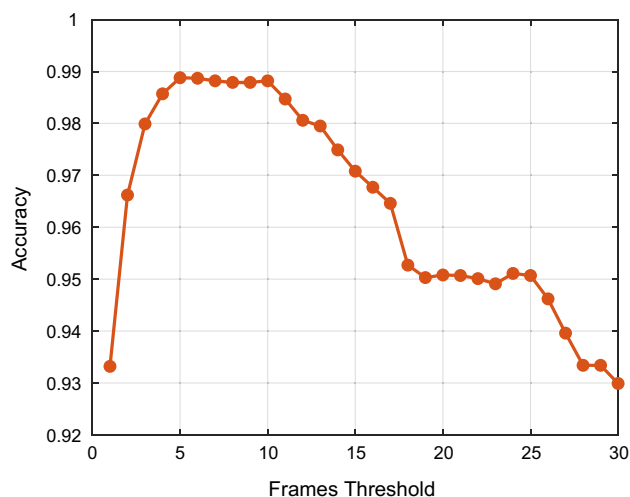


Fig. 5 Threshold value selection for Hockey-Fight dataset based on maximum test accuracy. The x -axis represents the threshold value (in a number of frames) starting from 0 to the length of video (for $f_N = 30$), whereas the y -axis represents accuracy result for that specific threshold value

5.1 Datasets

The proposed model is evaluated on four standard benchmarking datasets: Hockey Fights [1], Movies [1], Crowd Violence [2] and BEHAVE [48] which are most related to violence or fighting category. These datasets provide a broad range of violent activity scenarios, including person-to-person fighting scenes, violence activities involving more than two persons and crowd violence with different camera angles and movements. The datasets have different illumination conditions and view perspectives. Table 1 summarizes different statistics on all four datasets.

5.1.1 Hockey fight dataset

This dataset consists of video clips of ice hockey matches from National Hockey League (NHL). The dataset comprises of total 500 fighting and 500 non-fighting video clips. Each video has a frame-rate of 25fps with the duration of around 2 seconds (50-frames). The resolution of each frame is 360×280 .

5.1.2 Movies dataset

The second dataset also introduced in [1] involves clips from different action movies comprising of fights or non-fight actions. The video clips, which include violent or fight scene, are taken from different action movie sequences, whereas the non-fight clips are taken from random action recognition datasets such as UCF-101. This dataset contains 100 videos for both fights and non-fight labels. In contrast to hockey fight dataset, these videos consist of different context, illumination conditions and different backgrounds as videos are short clips taken from different types of movies.

5.1.3 Crowd violence dataset

The third dataset “Violent-Flows” introduced in [2] contains violent activities of real-world crowds. The clips are made from YouTube videos, in which violent activities include different scenarios of crowd fights. This dataset contains total of 246 videos with a frame resolution of 320×240 pixels. Each video is 1 to 6 s in duration.

5.1.4 BEHAVE dataset

This dataset [48] contains ten different activities performed by a group of people, such as running, chasing, following a person and fighting. The dataset contains four long videos with two different camera positions and locations with a total of around 200,000 frames, each of size 640×480 . Each video clip includes several activities. For the experimental purposes, we choose 20 clips involving fighting action and 50 clips of non-fighting actions (other activities) from each of the four videos to create a subset which includes total 280 clips with 80 fights and 200 non-fight scenes.

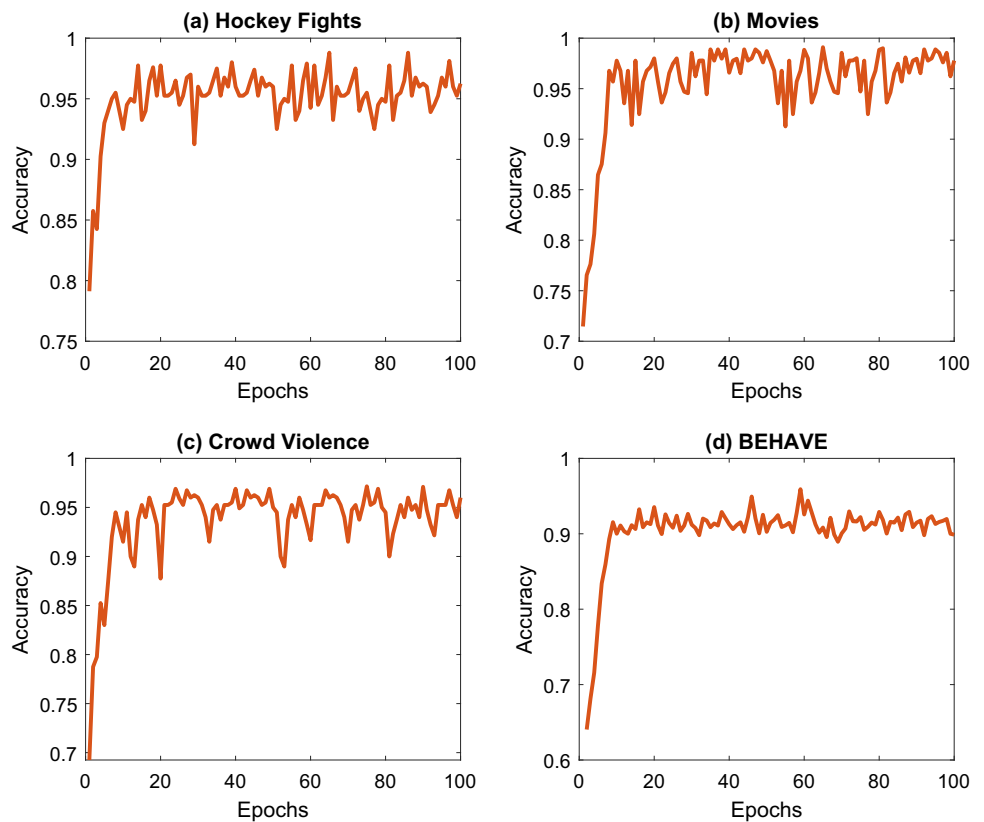
5.2 Results and discussion

The accuracy of the proposed technique for *HockeyFight*, *Movies* and *ViolentFlow* datasets is evaluated using fivefold cross-validation method. Each dataset is divided into five equal size partitions randomly, and four folds are used for training, and one is used for testing. The model is trained for

Table 1 Statistics summary for all four dataset used for the evaluation of proposed method

Dataset	Frames size	Total no. of videos	Fights scenes		Non-Fights scenes	
			No. of videos	Frame rate	No. of videos	Frame rate
Hockey Fights [1]	360×280	1000	500	25	500	25
Movies [1]	360×250	200	100	25	100	29.9
Crowd Violence [2]	320×240	246	123	25	123	25
BEHAVE [48]	640×480	280	80	25	200	25

Fig. 6 Mean test fold accuracy graphs for four datasets. **a** Maximum testing accuracy is 98.8% for Hockey Fight dataset **b** Maximum testing accuracy is 99.1% on Movies dataset **c** Maximum testing accuracy is 97.1% on Crowd Violence dataset and **d** Maximum testing Accuracy on BEHAVE dataset is 95.9%



each of these four folds and maximum accuracy is obtained for each of the test folds. Out of these five maximum test accuracies, the mean and standard deviation is calculated to evaluate the model. However, due to imbalanced *BEHAVE* dataset, we use stratified³ fourfold cross-validation method where 3 folds are used for training and 1 is used for testing. The stratified cross-validation method ensures approximately the same proportion of both classes (e.g. fight and non-fight) in each of the randomized shuffled folds.

The network parameters are selected on the basis of maximum test performance accuracy for each dataset. For each dataset, the network is run for a total of 100 epochs. Fig. 6 show the means test folds accuracy curves for each dataset.

For HockeyFight dataset the maximum performance accuracy achieved by the proposed model is 98.8% with a loss value of 0.0386 which occurs first on the 65th epoch. The best accuracy of 99.1% is obtained for Movies dataset at epoch 81 with a loss value 0.0179. The test accuracy on Crowd Violence dataset is 97.1% which is occurred first at epoch 91 with loss value of 0.04716, whereas for the BEHAVE dataset, for which only fighting category is used in this proposed method, gives the accuracy of 95.9% at epoch 60. We selected the initial learning rate to a larger value in the range of 10^{-2} to 10^{-3} , as at the start of the training the random initialized weights are

not close to optimal position. The learning rate is gradually decreased after each epoch, as the large initial learning rate can cause sub-optimal convergence of the model. The model is tested on different learning rates and in Table 2, two best accuracies and loss values obtained for a particular learning rate are shown. we observed that larger initial learning rates can lead to poor loss function optimization. As for hockey fight dataset, the model achieved the minimum loss value 7.85×10^{-1} with accuracy of only 83.1%. The model performs best at learning rate of 10^{-5} for HockeyFight, Crowd violence and BEHAVE whereas for Movies dataset the best minimum loss is achieved with a learning rate of 10^{-4} . Figure 7 shows the average values of training and test losses for four datasets over all cross-validation folds.

We also visualize the t-SNE-based feature embeddings [51] for our proposed model using HokeyFight dataset. The dimensions of features, obtained from the output the model, are reduced to 2. In Fig. 8, we can see that the features associated with same classes (violent and Non-violent) are clustered together while there is also a more spread out pattern.

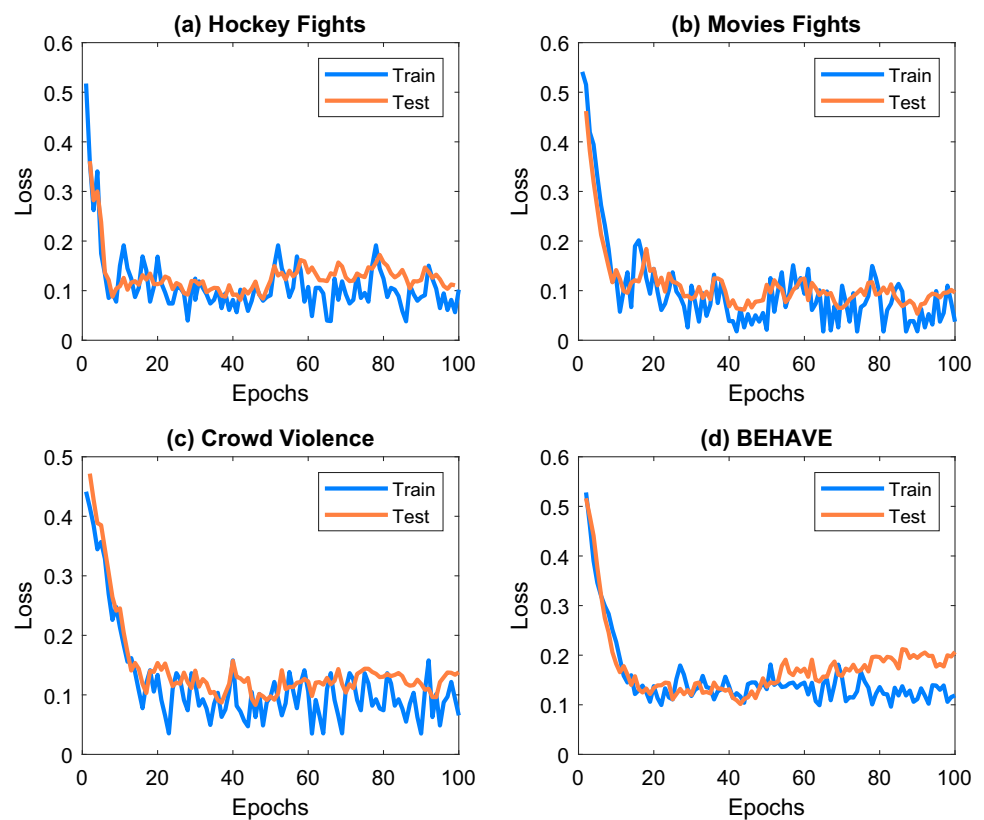
In Fig. 9 we presented the qualitative results in the form of false-positive and true-positive detection of our framework for four datasets. The quantitative results in the form of confusion matrices for each dataset are shown in Fig. 10 which are obtained after summing the confusion matrices

³ StratifiedShuffleSplit and StratifiedKFold classes are used from scikit learn library (<https://scikit-learn.org>).

Table 2 Statistics analysis for model convergence for different learning rates. The epoch where the accuracy is maximum for a particular learning rate is shown in the table

Dataset	Initial Learning rate	Epoch withmax ACC	ModelAccuracy (%)	Loss
Hockey Fights	10^{-2}	55	79.7	9.74×10^{-1}
		69	83.1%	7.85×10^{-1}
Movies	10^{-5}	65	98.8	3.86×10^{-2}
		86	98.0	5.91×10^{-2}
	10^{-3}	37	69.5	3.165×10^{-1}
		79	75.6	4.184×10^{-1}
Crowd Violence	10^{-4}	65	98.99	1.758×10^{-3}
		81	99.1	1.79×10^{-3}
	10^{-3}	59	81.8	3.524×10^{-2}
		91	82.1	4.71×10^{-2}
BEHAVE	10^{-5}	75	97.0	3.524×10^{-2}
		90	97.1	4.716×10^{-2}
	10^{-3}	54	64.8%	9.813×10^{-1}
		97	66.7	7.484×10^{-1}
	10^{-5}	46	94.8	5.991×10^{-2}
		60	95.9	5.097×10^{-2}

Fig. 7 Mean training and test loss curves for four datasets over all cross-validation folds **a** For Hockey Fight **b** Movies **c** Crowd Violence **d** BEHAVE



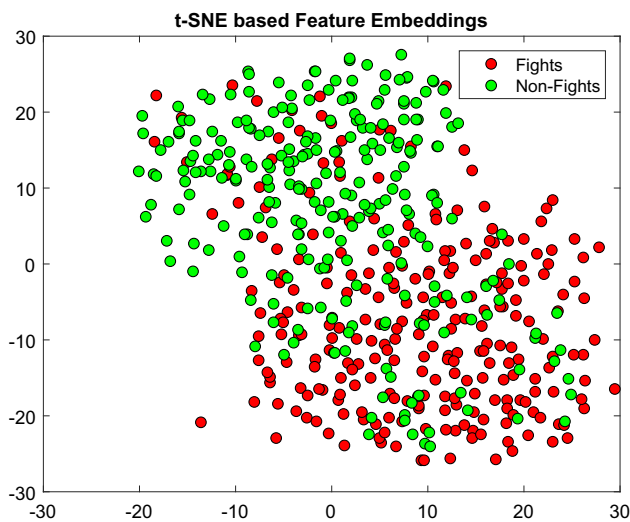


Fig. 8 Visualization of t-SNE-based feature embeddings of proposed model using HockeyFight dataset

from each test fold. The Receiver Operating Characteristic (ROC) curves for each dataset are obtained to further evaluate the true positive and false positive rate of our model for each dataset. Fig. 11 shows the ROC curves and AUC value for each dataset.

5.3 Comparative analysis

In this section we present the detailed comparison of our proposed model with various existing state-of-the-art methods. We compare our model with both deep learning and handcrafted-based approaches. The handcrafted-based approaches includes: Violent Flow descriptor (ViF) [2] where the authors proposed violent flow descriptor and used SVM

for final classification prediction. Gao et al. [22] proposed Oriented Violent Flow (OVIF) which is based on ViF descriptor with motion magnitudes and orientations between flow vectors. Nievas et al. [1] combined different feature descriptors for action recognition such as Motion Scale invariant Feature Transform (MoSIFT), Histogram Intersection Kernel (HIK) and Histogram of Oriented Flow (HOF) and followed the Bag of Words (BoW) approach for the purpose of violence detection. Mohammadi et al. [52] used behavioral heuristics and proposed a new framework named Visual Information Processing Signature (VIPS) while following the BoW approach. Xu et al. (MoSIFT+KDE+SC) [53] proposed violence detection method by combining sparse coding method with MoSIFT while using Kernel Density Estimation (KDE) for feature selection. Gracia et al. [12] used motion blob detection for feature extraction where these feature are used to distinguish between violence and non-violence sequences. Deniz et al. [54] incorporated acceleration patterns by computing power transforms of two consecutive video frames to detect sudden motion changes. Zhang et al. [13] improved the Weber Local descriptor by incorporating local motion and low-level appearance information (MoI-WLD) and proposed a sparse model for dictionary to classify violent actions. Senst et al. [37] proposed a spatio-temporal model based on Lagrangian fields. The model used SIFT [18] and BoW (LaSIFT+BoW) approaches for motion detection. The deep learning based methods used in comparative analysis includes: Stacked Denoising Autoencoder (AMDN) [28], ThreeStream+LSTM [55], ConvLSTM [36], and BiConvLSTM [35].

Most of these methods including our proposed method used binary classification-based approach for violence detection which only detects the presence of violence in videos

Fig. 9 **a** True-positive and **b** False-positive (right) detections of our proposed framework. Examples are shown from the HockeyFight (first row), the Movies (second row), Crowd Violence (third row) and the BEHAVE (fourth row) datasets

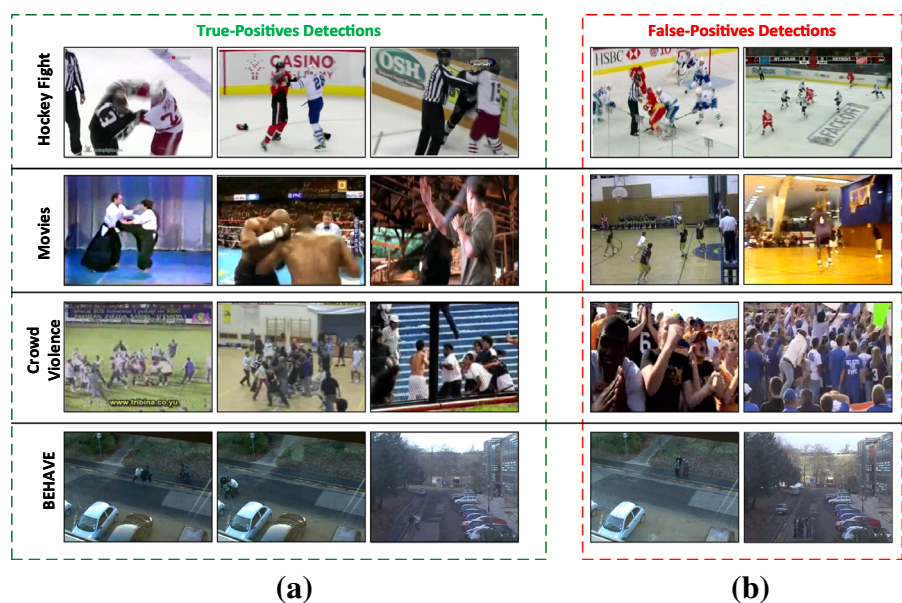
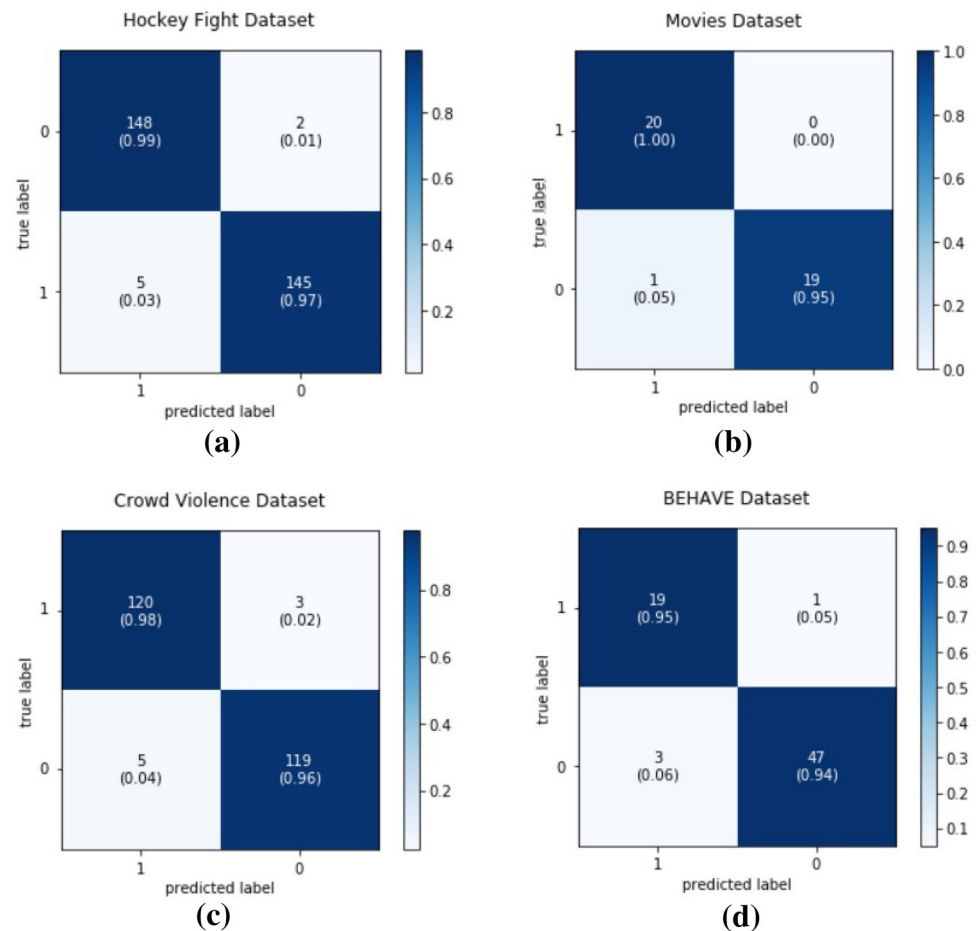


Fig. 10 Confusion Matrices for **a** HockeyFight **b** Movies **c** CrowdViolence and **d** BEHAVE dataset. [Label 1 : Fight (Positive) , Label 0 : Non fight (Negative)]



and results a binary-class (violence / non-violence) prediction output. Our methods is different from multi-class action classification problem such as proposed in [3,4] where the model can classify different classes of action. Our model is also different from anomaly detection-based methods [28,56] where the model is trained only on normal events and the anomaly is detected based on outlier detection.

Our method is most closely related to Sudhakaran and Lanz. [36] and Hanson et al. [35] where the former approach used pre-trained CNN to extract the spatial features from sequential input frames and ConvLSTMs are used for learning temporal dependencies. After all the input frames are applied to network, then the prediction is made through the fully connected layer. Hanson et al. [35] followed the similar architecture but used Bi-directional LSTMs instead of simple LSTMs. Both these methods are trained in supervised binary classification manner and results the prediction at video level.

Following the intuition from these methods [35,36] we proposed a model which also incorporates CNN and LSTMs for both spatial and temporal feature learning. However, different from these approaches our proposed model is trained on two-sequential frames and from these sequential frames input, we incorporated the motion information through the

fusion of multi-level spatial features. We also proposed wide-dense residual block for learning both combined low-level and high-level fused spatial features, which results in better accuracy results. Moreover, our proposed method can result in the prediction for each frame of the video which is useful for the situations where a particular violent action in a video consists of very short duration. Table 3 shows the accuracy evaluation ($ACC \pm SD$) of proposed model and comparison of above discussed methods whereas Fig. 12 shows the ROC curve comparisons for HockeyFight and Crowd Violence datasets.

We also compared our method with several baseline models where the first model consists of pre-trained CNN based on AlexNet [23] for spatial feature extraction and the last fully connected layer of CNN which has total 1000 hidden units is connected to LSTM layer with same number of units for learning temporal features. The model is trained in the similar configuration as describe in [36] where the LSTM units learn the temporal features from all the input video frames.

The second model consist of CNN with additional residual blocks (RB) of same configuration as the proposed in [43] but LSTM unit is replaced with the fully connected(FC) layer

Fig. 11 ROC curve and AUC value comparison for each dataset. **a** For Hockey Fight **b** Movies **c** Crowd Violence **d** BEHAVE

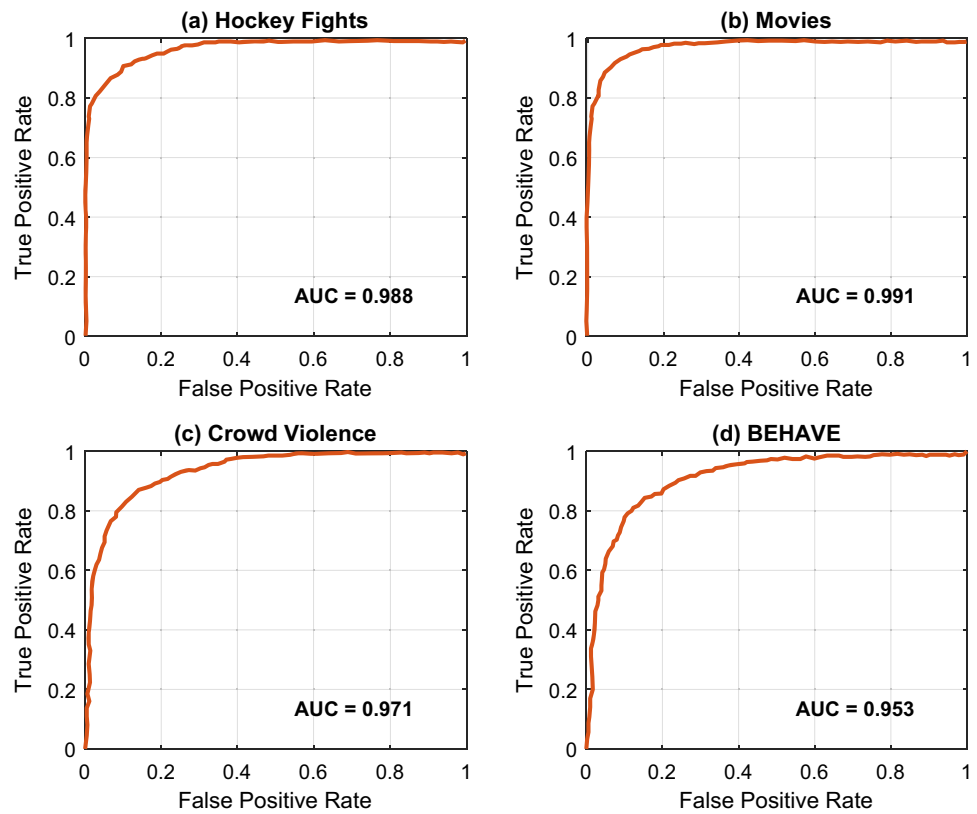


Table 3 The classification accuracy and standard deviation (ACC $\pm$ SD) comparison of the proposed model with the existing methods

Method	Datasets			
	Hockey Fights [1]	Movies [1]	Crowd Violence [2]	BEHAVE [48]
ViF [2]	82.9 $\pm$ 0.14%	–	81.3 $\pm$ 0.21%	83.62 $\pm$ 0.19%
OVIF [22]	87.5 $\pm$ 1.7%	–	88 $\pm$ 2.45%	–
MoSIFT+BoW [1]	88.8 $\pm$ 0.75%	–	83.42 $\pm$ 8.0%	81.65 $\pm$ 0.23%
MoSIFT+HIK [1]	90.9%	89.5%	–	–
VIPS [52]	–	96.91%	86.61	95.73%
HOG + BoW [1]	88.77 $\pm$ 0.73%	–	57.43 $\pm$ 0.37	58.97 $\pm$ 0.34%
MoSIFT+KDE+SC [53]	94.3 $\pm$ 1.68%	–	89.05 $\pm$ 3.26%	87.07 $\pm$ 0.13%
Gracia et al. [12]	82.4 $\pm$ 0.4%	97.8 $\pm$ 0.4%	–	–
Deniz et al. [54]	90.1 $\pm$ 0%	98.0 $\pm$ 0.22%	–	–
MoI-WLD [13]	96.8 $\pm$ 1.04%	–	93.19 $\pm$ 0.12%	88.83 $\pm$ 0.11%
LaSIFT +BoW [37]	94.42 $\pm$ 2.82%	94.95 $\pm$ 4.57%	93.12 $\pm$ 8.77%	–
AMDN [28]	89.7 $\pm$ 1.13%	–	84.72 $\pm$ 0.17%	84.22 $\pm$ 0.17%
Three streams +LSTM [55]	93.9%	–	–	–
ConvLSTM [36]	97.1 $\pm$ 0.55%	100$\pm$0%	94.57 $\pm$ 2.34%	–
BiConvLSTM [35] (Spatiotemporal model)	97.9 $\pm$ 0.55%	100$\pm$0%	96.32 $\pm$ 1.52%	–
Proposed Method	98.8$\pm$0.5%	99.1$\pm$0.3%	97.1$\pm$1.7%	95.9$\pm$2.3%

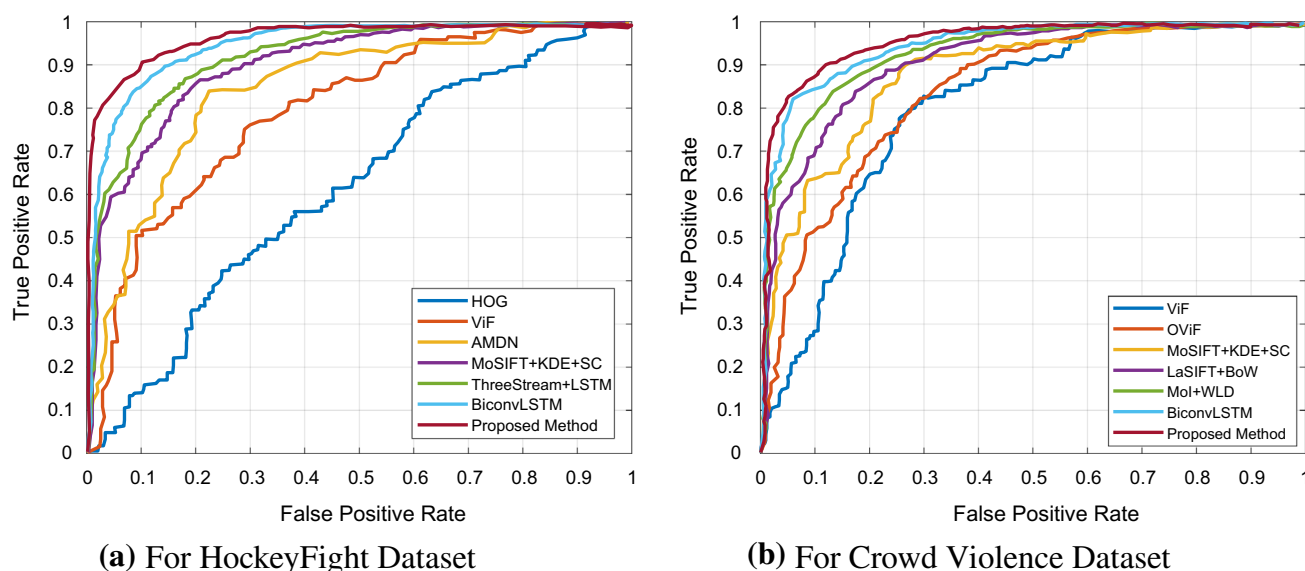


Fig. 12 ROC curve comparison of proposed method with existing techniques. **a** HockeyFight dataset. **b** Crowd Violence dataset

Table 4 Comparison with baseline models for HockeyFight dataset

Model	Frame-level Accuracy	Video-level Accuracy (%)
CNN+LSTM	—	93.8
CNN+RB+FC	92.51%	94.7
CNN+FF+RB+LSTM	97.2%	98.2
CNN+FF+WDRB+LSTM (proposed)	97.9%	98.8%

for classification. Also this model does not use spatial feature fusion and the model is trained for two sequential adjacent frames. The third model is similar to the proposed model. However, the residual block is used in default configuration [43] as shown in Fig. 4b.

These models are trained and tested on HockeyFight dataset. We have observed that the first model (CNN+LSTM) provide video-level accuracy result of only 93.8%. However, the second model (CNN+RB+FC), with the inclusion of simple residual layer block which is trained on two sequential frames provide better accuracy results of 94.7% video level accuracy results but relatively mediocre results at frame-level 92.51%. The third model (CNN+FF+RB+LSTM) which is based on feature fusion, residual layer block [43] and LSTM provides the video level accuracy of 98.2% whereas the proposed model with FF, WDRB provide frame-level accuracy of 97.9% and video-level accuracy of 98.8% that show that wide dense residual block(WDRB) and multi-level feature fusion in proposed model leads to better performance improvement. Table 4 shows the results of baseline models compared to proposed model.

6 Conclusion

This paper has proposed a novel method for detecting violent activities in videos by learning fused spatial feature maps of two sequential input frames using VGG-16 and additional wide-dense residual layer blocks. The temporal features have been learned using Long Short-Term Memory units. The transfer learning using a pre-trained model on the ImageNet dataset has considerably enhanced the performance and reduced the training time. The use of additional residual blocks have further improved performance accuracy. The LSTM blocks have learned the temporal dependencies from combined features of two input frames to differentiate normal actions from violent actions. The model can predict the classification result at frame-level. The proposed method has been tested on four standard benchmarking datasets and experimental results have demonstrated that the proposed method is effective for numerous types of fighting scenes including crowd violence activities, and provides a significant superior performance compared to the state-of-the-art violence detection methods.

Acknowledgements This research is partly supported by NSFC, China (No: 61876107, U1803261) and Committee of Science and Technology, Shanghai, China (No. 19510711200).

Compliance with ethical standards

Conflict of interest The authors declare that we have no conflict of interest.

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